

Solenoid Valve

Lecture 14

Solenoid Valve

- A solenoid valve is an electromechanical device used to control the flow of a liquid or gas. It is comprised of two features: a solenoid and a valve. The solenoid is an electric coil with free-moving ferromagnetic material in the center of the coil, often referred to as a “plunger.” When voltage is applied to the coil, the solenoid is energized. This action creates a magnetic field that either attracts or repels the plunger and causes it to translate linearly (see Figure 1). The plunger either opens or closes the orifice the media flows through.
- The movement of the plunger alters the position of the components within the valve body to control the specific flow, direction, and pressure of fluid moving through the valve. When current or voltage is removed from the coil, the magnetic field collapses and all internal elements return to their de-energized position.

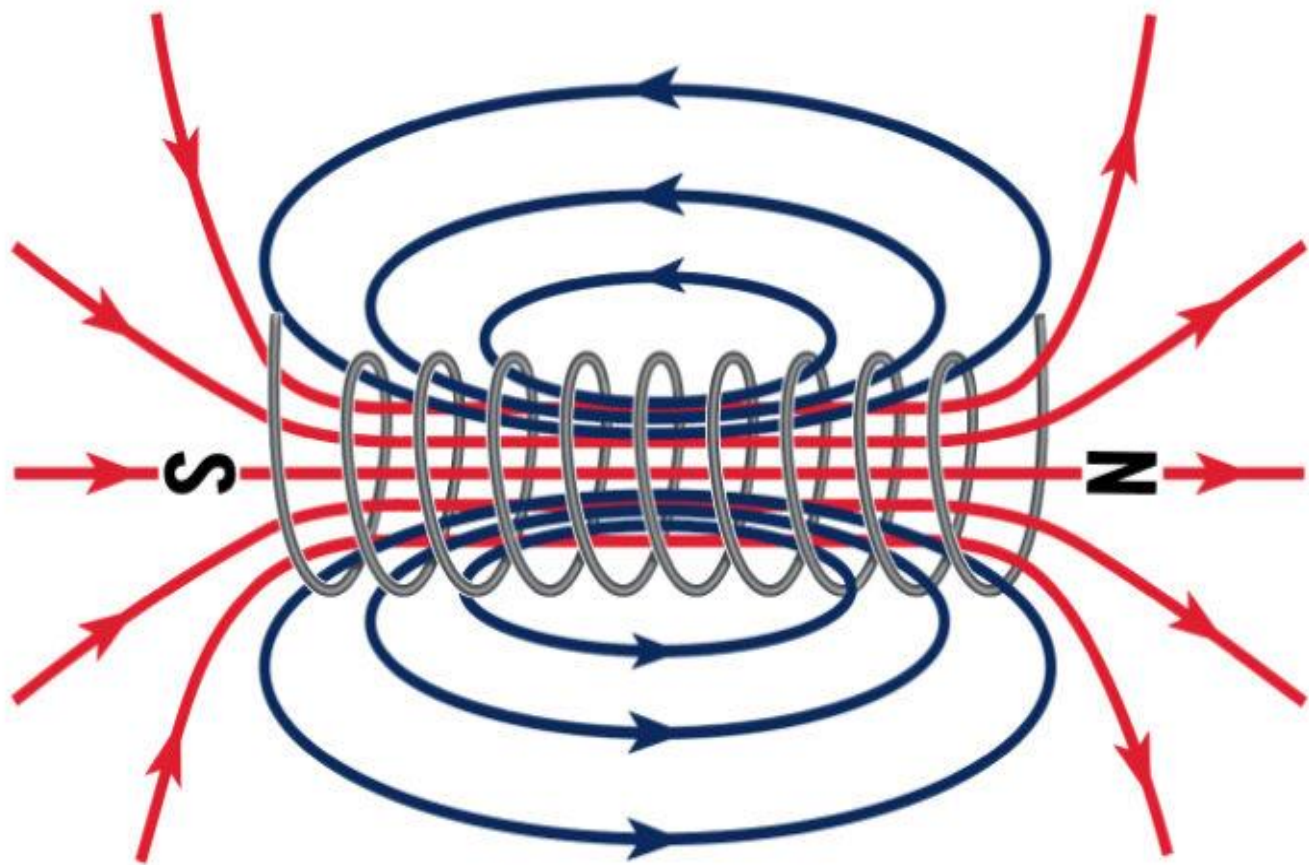


Figure 1: Electrical Field Through Solenoid Coil

- Different designs, construction materials, and circuit functions allow solenoid valves to be used in various applications. Since they are electrically controlled, they can be controlled remotely and automatically. They are commonly found in water treatment, automotive, food processing, and other industrial applications.



Figure 2: 2/2-way solenoid valve

The work principle of a solenoid valve

- A solenoid valve consists of two main components: a solenoid and a valve body. A solenoid has an electromagnetically inductive coil around an iron core at the center called the plunger. AC coils have a shading ring, which prevents vibration and humming.
- At rest, the valve can be normally open or normally closed. In the de-energized state, a normally closed valve is closed. When current flows through the solenoid, the coil is energized, creating a magnetic field. This creates a magnetic attraction with the plunger, moving it and overcoming the spring force. The plunger lifts so the seal opens the orifice, allowing the media to flow through the valve. A normally open solenoid valve works oppositely.

Components of a solenoid valve

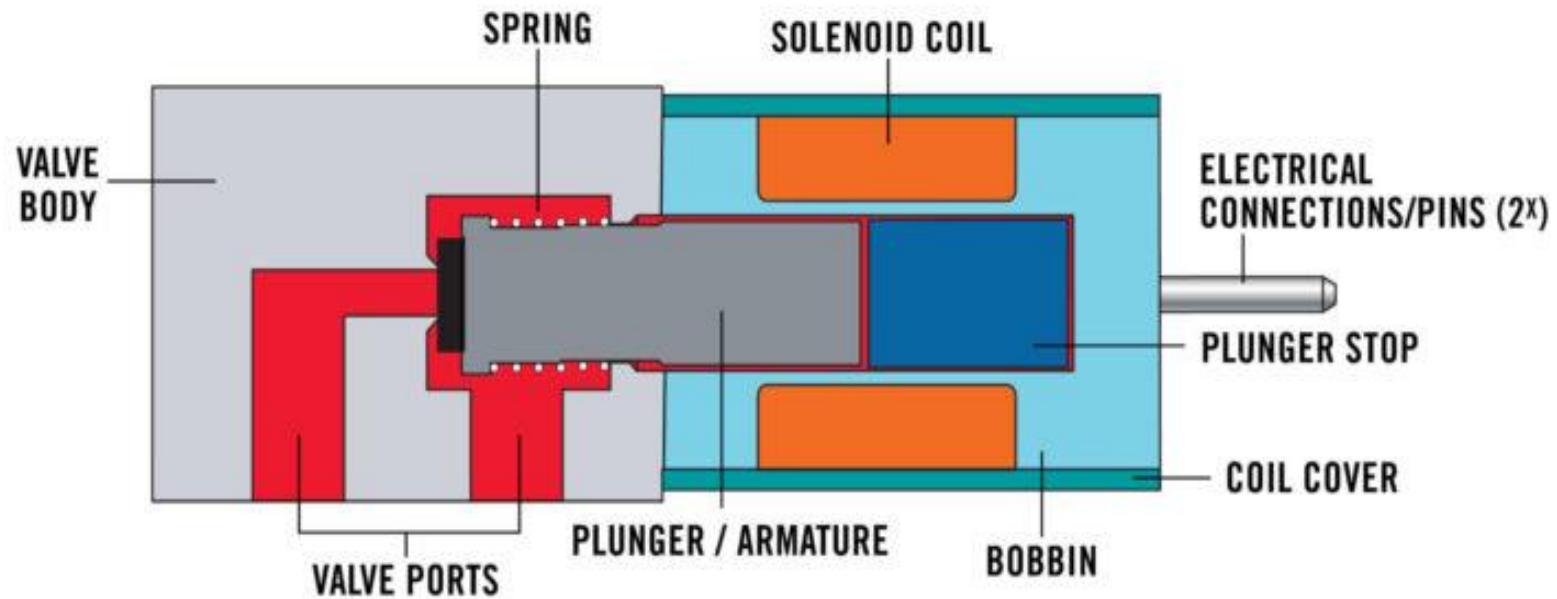


Figure 3: Components of a solenoid valve

Coil: The coil is commonly made from insulated copper wire. When a current is applied to the coil, an electromagnetic field forms around the cross-section.

Electrical Connection: The coil is connected to an interface with an electrical drive circuit and power source. Each coil typically has two connections: one to the power source and one to ground.

Bobbin: The coil wire is commonly wrapped around a structural piece called a bobbin. The bobbin helps to hold the wire in the desired position to create a specific magnetic field.

Coil Cover/Shielding: A cover is placed around the solenoid coil for protection. The cover may be constructed from a material that will provide shielding. Shielding protects the solenoid coil from interference from external electrical noise and protects the surrounding equipment from interference caused by electromagnetic radiation from the solenoid coil.

- . **Plunger/Armature:** A ferromagnetic plunger (sometimes referred to as an armature) is placed in the center of the coil or bobbin to convert electrical energy into mechanical work. The magnetic field created from the energized coil will attract or repel the plunger coaxially. The plunger is typically manufactured from a soft iron. Stainless steel materials are not suitable for the plunger because they do not become demagnetized after the current is removed and would not return to their original state.
- . **Valve Seal:** The position and movement of the plunger controls the components of the valve body to seal or open flow paths.

- **Valve Body:** At a minimum, the valve body includes the valve seat and fluidic ports. In more complex valve designs, the valve body may also house additional sub-components to optimize valve performance.
- **Spring:** A spring is commonly used within the valve body to bias the plunger and other movable valve components into position when de-energized. Some designs incorporate multiple springs.
- **Plunger Stop:** A plunger stop is often employed to control the length of plunger movement, or stroke, when energized.

Solenoid valve types

- **1. Direct acting:**
- A direct-acting solenoid valve uses the solenoid to open or close without differential pressure.
- These valves are often used to control the flow of gas or liquid in a system.
- Direct-acting solenoid valves have the fastest operation, are reliable, and have a compact design.

2. Indirect acting:

Indirect-acting solenoid valves, whether servo-operated or pilot-operated, operate using pressure differences.

They require a minimum pressure differential of about 0.5 bar.

These valves include a diaphragm with a small hole that allows flow from the inlet to the outlet when the solenoid is energized and the pressure drops.

This system amplifies the pressure, allowing a small solenoid to control a large flow rate.

Indirect solenoid valves are used in applications with sufficient pressure differential and high desired flow rates, and they only allow media flow in one direction.

3. Semi-direct acting

Semi-direct acting solenoid valves merge the features of direct and indirect valves, enabling operation from zero bar while managing high flow rates.

These valves look like indirect valves with a movable membrane, a small orifice, and pressure chambers on both sides, but the solenoid plunger is directly connected to the membrane.

When the plunger lifts, it directly opens the valve and a second orifice, causing the pressure to drop and the membrane to lift.

This results in a valve operating from zero bar and managing significant flow rates.

These semi-direct operated valves, also known as assisted-lift solenoid valves, often have more powerful coils than indirect-operated valves.

Normally open and normally closed solenoid valves

- **Normally open solenoid valve:**
- A normally open (NO) solenoid valve is open when de-energized, allowing the media to flow through it.
- When current is sent to the coil, it creates an electromagnetic field that forces the plunger downwards, overcoming the spring force. The seal sits in the orifice and closes it, preventing media from flowing through the valve.
- Figure 4 shows the operating principle of a normally open solenoid valve in the de-energized and energized states.

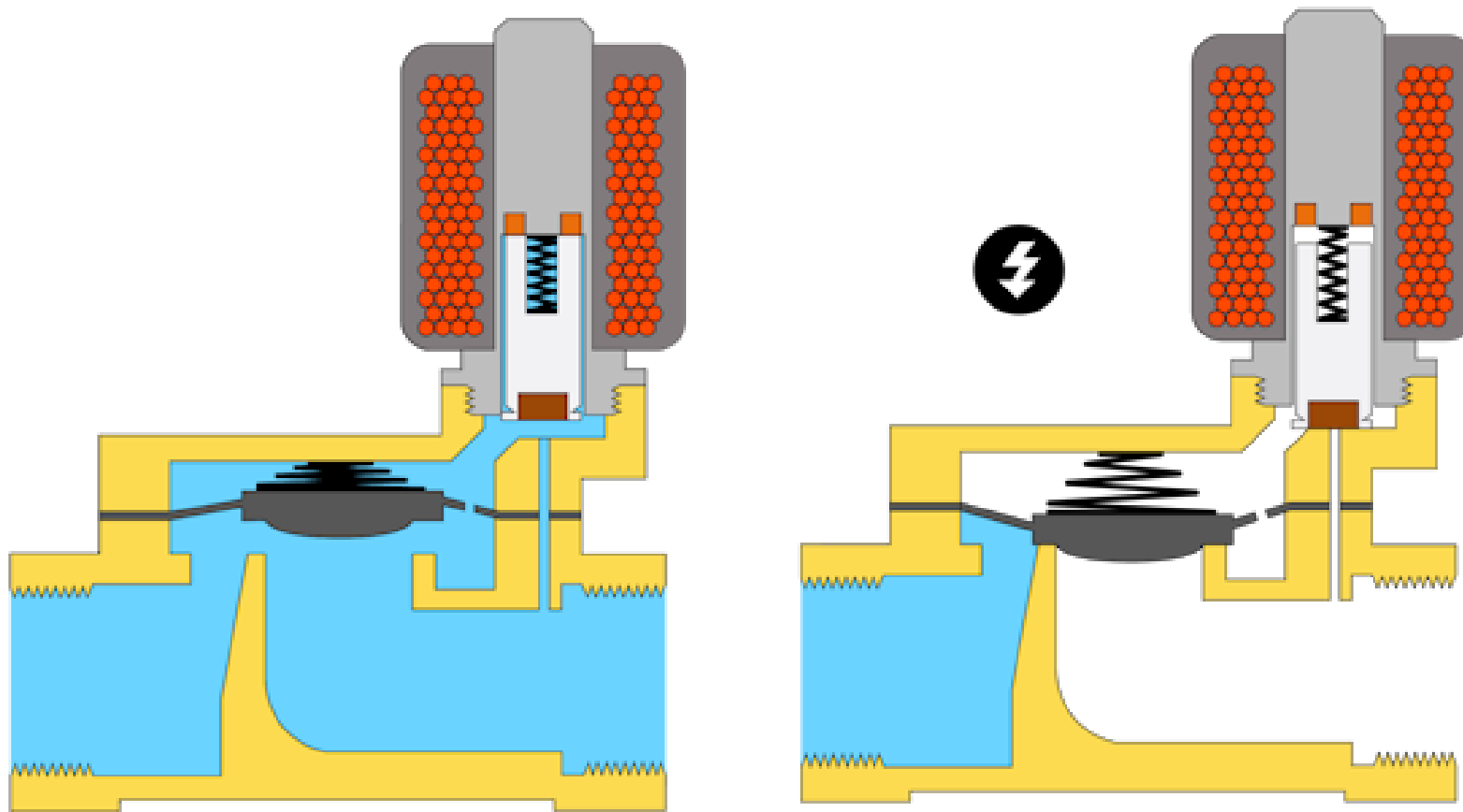


Figure 4: Operating principle of normally open solenoid valve: de-energized (left) and energized (right).

A normally open solenoid valve is ideal for applications that require the valve to be open for long periods of time, as this is more energy efficient. They can also be used for safety reasons if the application requires the valve to be open with no power (e.g., to prevent overpressure).

Normally closed solenoid valve

A normally closed (NC) solenoid valve is closed when de-energized, which prevents the media from flowing through it.

When current is sent to the coil, it creates an electromagnetic field that forces the plunger upwards, overcoming the spring force. This unseats the seal and opens the orifice, allowing the media to flow through the valve.

Figure 5 shows the operating principle of a normally closed solenoid valve in the de-energized and energized states

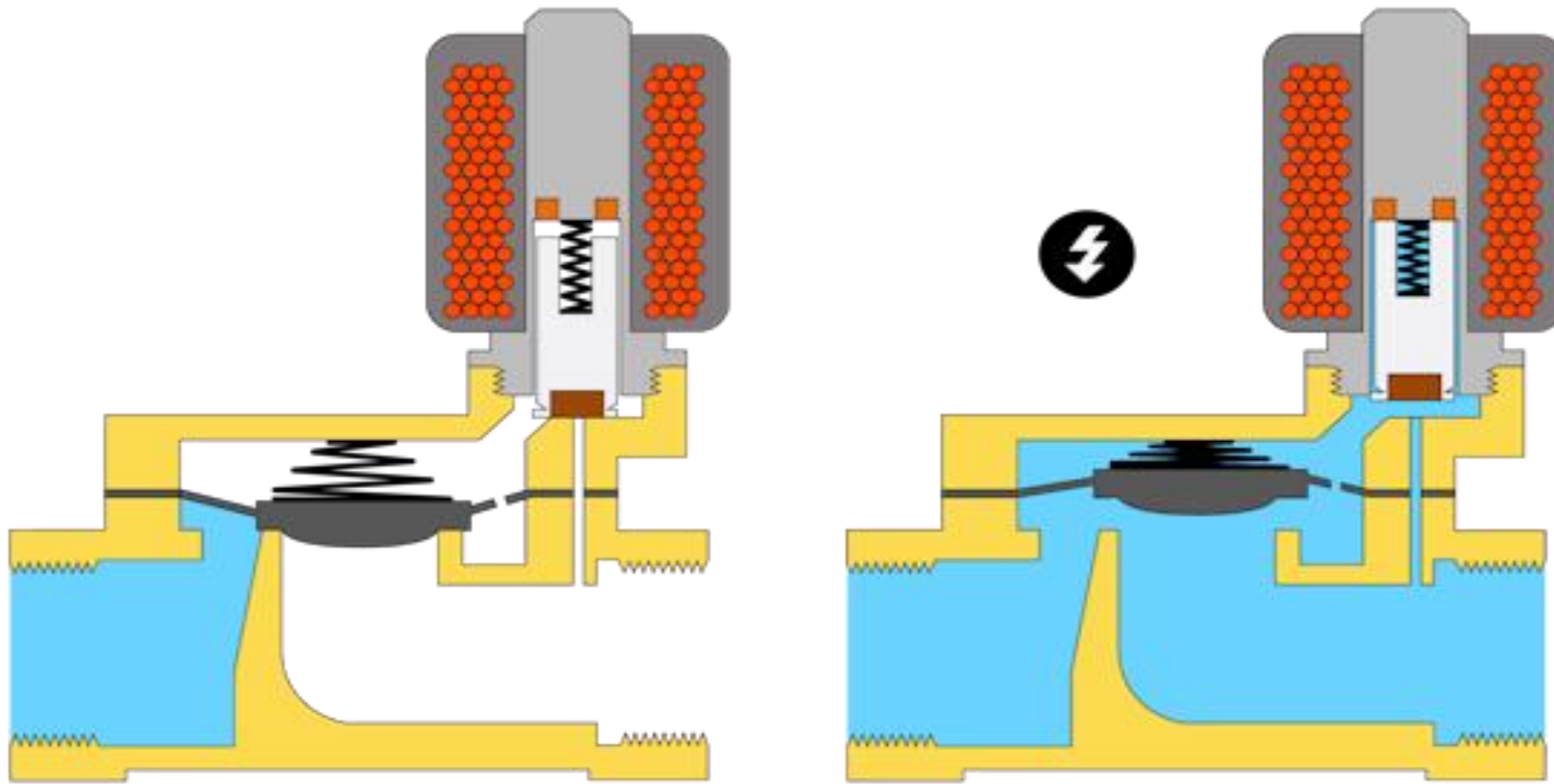


Figure 5: Operating principle of normally closed solenoid valve: de-energized (left) and energized (right).

A normally closed solenoid valve is ideal for applications that require the valve to be closed for long periods of time, as this is more energy efficient. They can also be used for safety reasons if the application requires the valve to be closed without power (e.g., gas appliances).

For example, a solenoid valve in a dishwasher will be in the closed position when the dishwasher is off in order to conserve water. When the dishwasher is turned on, the valve will open for the period of operation during which water needs to flow onto the kitchenware.

Bi-stable solenoid valve

A bi-stable or latching solenoid valve can be switched by a momentary power supply.

When de-energized, the valve stays in the position it switched to. Therefore, **it is not normally open or closed**, as it stays in the current position when no power is applied.

They accomplish this by using permanent magnets rather than a spring. This gives the benefit of reduced power consumption.

2-way solenoid valve

2-way solenoid valves have two ports, an inlet, and an outlet, and are used to allow or block flow.

The flow direction through the valve is critical to ensure proper operation. Typically, an arrow indicates the flow direction on the body of the valve.



Figure 6: Fluid Control 2 way solenoid valve type.

3-way solenoid valve

A 3-way solenoid valve has three ports, each serving a distinct purpose: one for the inlet, one for the outlet, and one depending on the valve's configuration and application (exhaust, return, another inlet, or another outlet).



Figure 7: Fluid Control 3 way solenoid valve type.

The common designations for these ports:

- ❖ **P (pressure) port or inlet port:** This is where the pressurized fluid or gas enters the valve.
- ❖ **A (actuator) port or outlet port:** This is the working port connected to the device or actuator that the valve is intended to control, such as a [single-acting pneumatic cylinder](#) or a process line.
- ❖ **E (exhaust) port or R (return) port:** When the valve is switched to the exhaust position, this port vents the fluid or gas from the actuator or device back to the atmosphere or a reservoir. Depending on the valve's configuration and application, this port might also serve as an alternative inlet or outlet in some systems.

The specific function of the ports depends on the state of the solenoid valve (energized or de-energized) and the design (normally closed or normally open).

- **Normally closed (NC):** When the valve is de-energized, the P port is closed, and no flow is allowed from the inlet to the outlet. The A port is typically connected to the E or R port, allowing the actuator to exhaust. When energized, the valve opens the flow from the P port to the A port and closes off the E or R port.
- **Normally open (NO):** When the valve is de-energized, the P port opens to the A port, allowing flow from the inlet to the outlet. The E or R port is closed. When energized, the valve switches to close the flow from P to A and opens the connection between A and E or R, allowing the actuator to exhaust.